

Last Time

Distributions

- Discrete Distributions

- Continuous Distributions

Relationships Between Distributions

QQ Plots

Introduction to Statistical Inference

The Central Limit Theorem

Sampling Distribution of $\bar{X}$

Today

Finishing Up Distributions

QQ Plots

Introduction to Statistical Inference

The Central Limit Theorem

Sampling Distribution of $\bar{X}$

Confidence Intervals

When σ^2 is Unknown

Relationships Among Distributions

Leemis Chart

- ▶ $X \sim \text{Ber}(p): \sum_{i=1}^N X_i \sim \text{Bin}(N, p)$
- ▶ $X \sim \text{Bin}(N, p): \lim_{N \rightarrow \infty, Np = \lambda} \text{Bin}(N, p) \sim \text{Pois}(\lambda)$
- ▶ $X \sim \text{Bin}(N, p): \lim_{N \rightarrow \infty} \text{Bin}(N, p) \sim \mathcal{N}(Np, Np(1 - p))$
- ▶ $Z \sim \mathcal{N}(0, 1): \sum_{i=1}^k Z_i^2 \sim \chi^2(k)$
- ▶ $X \sim t(\nu): X^2 \sim F(1, \nu)$
- ▶ $X \sim t(\nu): \lim_{\nu \rightarrow \infty} t(\nu) \sim \mathcal{N}(0, 1)$
- ▶ $X^2 \sim \chi^2(k): \lim_{k \rightarrow \infty} \chi^2(k) \sim \mathcal{N}(k, 2k)$
- ▶ $X_1^2 \sim \chi^2(n_1), X_2^2 \sim \chi^2(n_2): \frac{X_1^2/n_1}{X_2^2/n_2} \sim F(n_1, n_2)$

QQ Plot

- ▶ Plot the quantiles of the observed data (y axis) against the quantiles of the theoretical distribution (x axis).
- ▶ The plot points should fall on the identity line $y = x$.

R code

```
data <- read.csv("GSS_1980.csv",header=TRUE)
age <- data$AGE

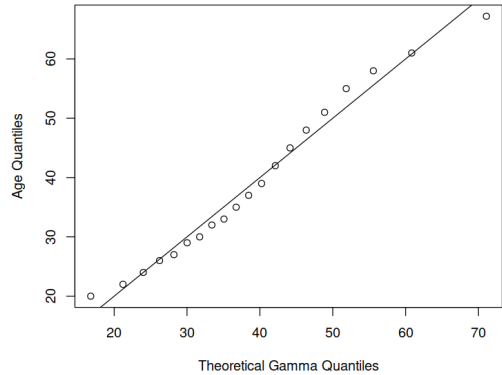
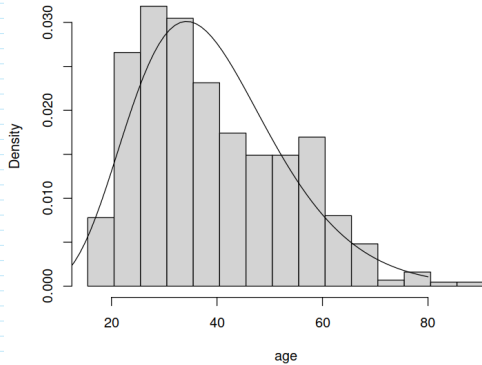
# Pick some good values for the gamma distribution
a <- mean(age)/var(age)
k <- mean(age)^2/var(age)

# Compute the quantiles for the data and the gamma model
obs <- quantile(age,seq(.025,.975,by=.05))
theory <- qgamma(seq(.025,.975,by=.05),k,a)

# Plot the data histogram with the gamma model
par(mfrow=c(1,2))
hist(age,freq=FALSE,breaks=seq(15.5,90.5,by=5),
      xlab="age",ylab="Density",main="")
lines(0:80,dgamma(0:80,k,a))

# QQ Plot
plot(theory,obs,type="p",ylab="Age Quantiles",
      xlab="Theoretical Gamma Quantiles")
abline(0,1)
```

QQ Plot



Inference: Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

Family	Sibling	
	Older	Younger
1	14	32
2	59	85
3	89	98
4	88	08
5	47	20
6	98	94
7	36	78
8	84	14
9	55	38
10	10	81
11	26	20
12	38	27
13	51	35
14	81	32
15	99	76

Inference: Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

- ▶ X = number of “+”s
- ▶ How is X distributed?

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
8	84	14	+
9	55	38	+
10	10	81	-
11	26	20	+
12	38	27	+
13	51	35	+
14	81	32	+
15	99	76	+

Inference: Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

- ▶ X = number of “+”s
- ▶ How is X distributed?
- ▶ $Pr(X \geq 10 | p = .5) = 0.15$

```
N <- 15  
p <- 0.5  
1 - pbinom(9, N, p)
```

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
8	84	14	+
9	55	38	+
10	10	81	-
11	26	20	+
12	38	27	+
13	51	35	+
14	81	32	+
15	99	76	+

Questions/Goals

- ▶ Scientific Question: Are older siblings more conservative than younger siblings?
- ▶ Statistical Question: Is the number of “+”s we observed likely to have occurred by chance?

Sampling Distributions

```
# Draw samples from an Exp(1) population
plot(seq(0,5,length=200),dexp(seq(0,5,length=200)),
      type='l',xlab = "Values of X",ylab="Density")

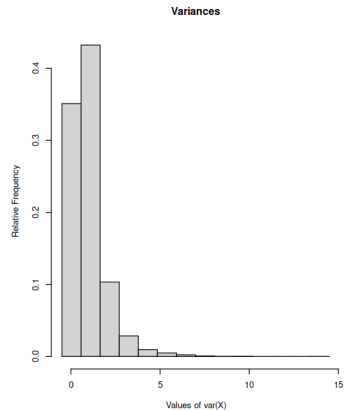
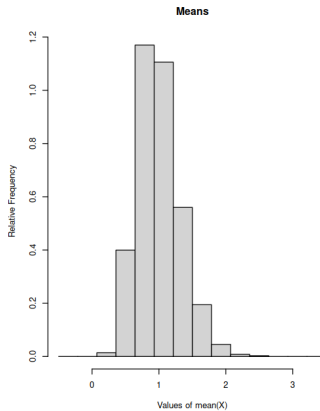
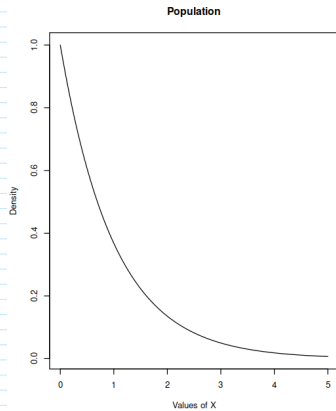
bignum <- 10000
N <- 10
means <- vector()
vars <- vector()

for (i in 1:bignum) {
  X <- rexp(N)
  means[i] <- mean(X)
  vars[i] <- var(X)
}

par(mfrow=c(1,2))
hist(means,breaks=seq(-.5,round(max(means))+.5,length=15),freq=FALSE,
      ylab="Relative Frequency",xlab="Values of mean(X)",main="Means")

hist(vars,breaks=seq(-.5,round(max(vars))+.5,length=15),freq=FALSE,
      ylab="Relative Frequency",xlab="Values of var(X)",
      main="Variances")
```

Sampling Distributions



Sampling Distribution

Definition (Sampling Distribution)

A *sampling distribution* is the distribution of possible values of $S(X)$ under all possible random samples of X of a fixed size N .

Distributions of Statistics

Single samples of size N

Parameter	Test Statistic	Distribution
μ	$\bar{X}$	$\mathcal{N}(\mu, \sigma^2/N)$
μ	$\frac{\bar{X} - \mu}{s/\sqrt{N}}$	$t(N-1)$
π	p	$\mathcal{N}(\pi, \pi(1-\pi)/N)$
σ^2	$\frac{(N-1)s^2}{\sigma^2}$	$\chi^2(N-1)$
ρ	$\frac{1}{2} \ln[(1+r)/(1-r)]$	$\mathcal{N}(\tanh^{-1}(\rho), 1/\sqrt{N-3})$

The Central Limit Theorem

Consider any variable X and a sample of size N of that variable.

As N gets large (goes to infinity), the shape of the distribution of the sample mean $\bar{X}$ will become normal.

The Central Limit Theorem

Consider any variable X and a sample of size N of that variable.

As N gets large (goes to infinity), the shape of the distribution of the sample mean $\bar{X}$ will become normal.

Formally: For an iid sample $X = \{X_1, X_2, \dots, X_N\}$ of size N drawn from a population with mean μ and standard deviation σ , define the random variable $S_N = \sum_{i=1}^N X_i$.

$$\lim_{N \rightarrow \infty} Pr\left(\frac{S_N - N\mu}{\sigma\sqrt{N}} \leq x\right) = P(Z \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz.$$

The Central Limit Theorem

Consider any variable X and a sample of size N of that variable.

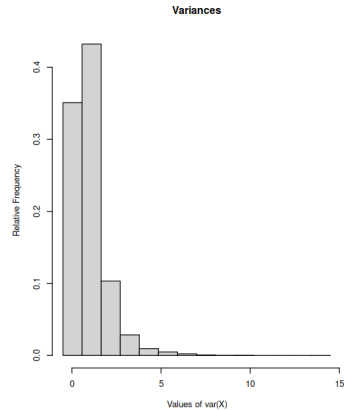
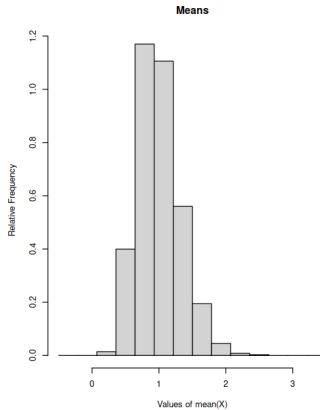
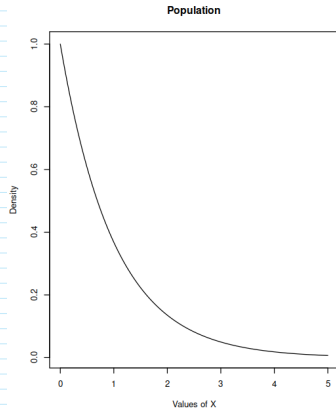
As N gets large (goes to infinity), the shape of the distribution of the sample mean $\bar{X}$ will become normal.

Formally: For an iid sample $X = \{X_1, X_2, \dots, X_N\}$ of size N drawn from a population with mean μ and standard deviation σ , define the random variable $S_N = \sum_{i=1}^N X_i$.

$$\lim_{N \rightarrow \infty} Pr\left(\frac{S_N - N\mu}{\sigma\sqrt{N}} \leq x\right) = P(Z \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz.$$

Regardless of the shape of the distribution of X !

Exponential Population



Central Limit Theorem

Why is the CLT important?

Central Limit Theorem

Why is the CLT important?

- ▶ It is the foundation of statistical inference across many different disciplines, data types, and analytic techniques.
- ▶ Many statistical procedures rely on assumptions of normality, which the CLT guarantees.
- ▶ We know how $\bar{X}$ will be distributed even if we know nothing about the population from which X was sampled.
- ▶ We know that the mean of $\bar{X}$ is equal to the population mean μ , which is the thing we want to tell stories about.

Normal Random Variables

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

- ▶ $X + c \sim \mathcal{N}(\mu + c, \sigma^2)$

- ▶ $cX \sim \mathcal{N}(c\mu, \sigma^2)$

If X_1 and X_2 are independent and $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, then

- ▶ $X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

- ▶ $aX_1 + bX_2 \sim \mathcal{N}(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$

If $X = \{X_1, X_2, \dots, X_N\}$ is an iid sample with $X_i \sim \mathcal{N}(\mu, \sigma^2)$ then

- ▶ $\sum_{i=1}^N X_i \sim \mathcal{N}(N\mu, N\sigma^2)$

If some number of independent normal random variables are linearly transformed or combined, the result is also a normal random variable.

The Sampling Distribution of the Mean

By the CLT, we know $\bar{X} \sim \mathcal{N}(\mu_{\bar{X}}, \sigma_{\bar{X}}^2)$

What are $\mu_{\bar{X}}$ and $\sigma_{\bar{X}}^2$?

$\mu_{\bar{X}}$

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{1}{N} \sum_{i=1}^N X_i\right) \\ &= \frac{1}{N} E\left(\sum_{i=1}^N X_i\right) \\ &= \frac{1}{N} \left(\sum_{i=1}^N E(X_i)\right) \\ &= \frac{1}{N} \left(\sum_{i=1}^N \mu\right) \\ &= \frac{1}{N} N\mu = \mu. \end{aligned}$$

$$\sigma_X^2$$

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{N} \sum_{i=1}^N X_i\right) \\ &= \frac{1}{N^2} \text{Var}\left(\sum_{i=1}^N X_i\right) \\ &= \frac{1}{N^2} \left(\sum_{i=1}^N \text{Var}(X_i)\right) \\ &= \frac{1}{N^2} \left(\sum_{i=1}^N \sigma^2\right) \\ &= \frac{1}{N^2} N \sigma^2 = \frac{\sigma^2}{N}. \end{aligned}$$

Estimating μ

- ▶ We have $\bar{X}$; we want μ .
- ▶ We know that $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/N)$
 1. $\bar{X}$ is a point estimate of μ : $\bar{X} = \hat{\mu}$.
 2. Using what we know about the distribution of $\bar{X}$ we can construct an interval that *covers* μ with some selected probability.

Point Estimate

What's the best single, numerical guess for μ ?

- ▶ $\bar{X} = \hat{\mu}$
- ▶ Consistent: Law of Large Numbers: $\lim_{N \rightarrow \infty} \bar{X} = \mu$
- ▶ Unbiased: $E(\bar{X}) = \mu$
- ▶ Other good properties

Interval Estimate

- ▶ Assume we know what σ^2 is.
- ▶ An *interval estimate* is a range of possible values for μ .
- ▶ Based on the sampling distribution for $\bar{X}$, we can determine the width of an interval that will include μ some specified percentage of the time.

Back to the Normal Distribution

Consider a standard normal Z random variable.

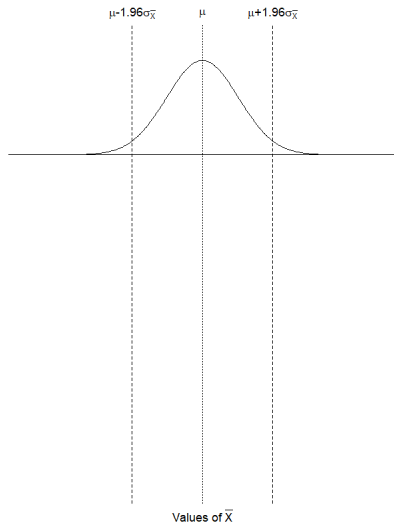
Back to the Normal Distribution

Consider a standard normal Z random variable.

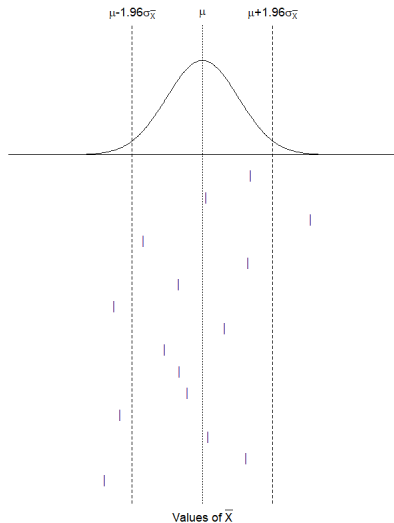
Place upper and lower bounds symmetrically around $\mu_Z = 0$.

Suppose we want the probability of sampling a z -score between those boundaries to be 95%. What are the values of the boundaries?

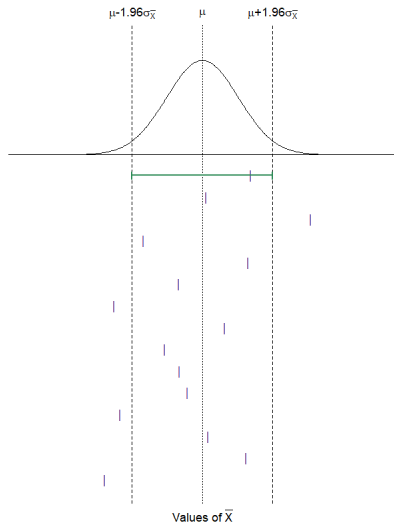
Understanding Confidence Intervals



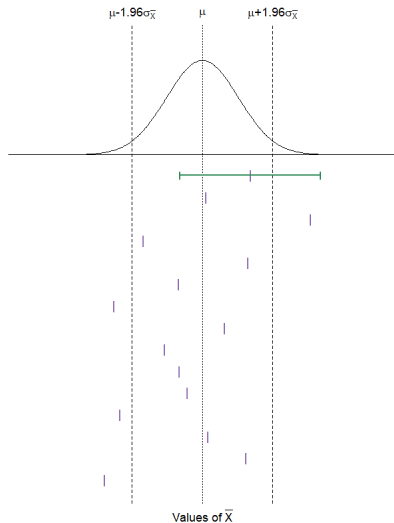
Understanding Confidence Intervals



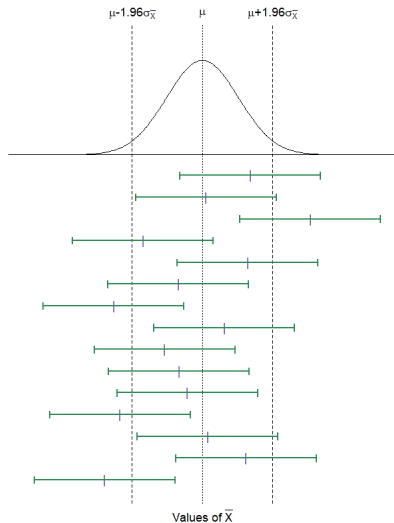
Understanding Confidence Intervals



Understanding Confidence Intervals



Understanding Confidence Intervals



Variability

The interval $\bar{X} \pm 1.96\sigma_{\bar{X}}$ will cover μ for 95% of the values of $\bar{X}$ that we observe.
What is random?

- ▶ $\bar{X}$
- ▶ σ
- ▶ μ
- ▶ Interval $[L, U]$
- ▶ Interval width $U - L$

Another Way To Think About It

How do we go from an interval around $Z_{\bar{X}}$ to an interval around $\bar{X}$?

Another Way To Think About It

How do we go from an interval around $Z_{\bar{X}}$ to an interval around $\bar{X}$?
Start with what we know:

$$\begin{aligned} Pr(-1.96 < Z_{\bar{X}} < 1.96) \\ = .95 \end{aligned}$$

Another Way To Think About It

How do we go from an interval around $Z_{\bar{X}}$ to an interval around $\bar{X}$?
Start with what we know:

$$\begin{aligned} & Pr(-1.96 < Z_{\bar{X}} < 1.96) \\ &= .95 \\ &= Pr\left(-1.96 < \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} < 1.96\right) \\ &= Pr(-1.96\sigma_{\bar{X}} < \bar{X} - \mu < 1.96\sigma_{\bar{X}}) \\ &= Pr(-\bar{X} - 1.96\sigma_{\bar{X}} < -\mu < -\bar{X} + 1.96\sigma_{\bar{X}}) \\ &= Pr(\bar{X} + 1.96\sigma_{\bar{X}} > \mu > \bar{X} - 1.96\sigma_{\bar{X}}) \\ &= Pr(\bar{X} - 1.96\sigma_{\bar{X}} < \mu < \bar{X} + 1.96\sigma_{\bar{X}}) \end{aligned}$$

σ^2 is Unknown

$$T = \frac{\bar{X} - \mu}{s/\sqrt{N}} \sim t(N-1)$$

Z-interval:

$$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{N}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{N}} \right]$$

T-interval:

$$\left[\bar{X} - t_{\alpha/2} \frac{s}{\sqrt{N}}, \bar{X} + t_{\alpha/2} \frac{s}{\sqrt{N}} \right]$$

